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Palaeomagnetism of the Jotnian lavas and sediments and post-Jotnian dolerites of central Scandinavia

JOHN D. A. PIPER and ROBERT L. SMITH

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Alternating field and thermal demagnetisation studies of lavas and sediments assigned to the Proterozoic Jotnian of western Sweden (Dalarna) and eastern Norway are reported. Hematite is the predominant remanence carrier in the sediments. These locally record a dipolar axis of magnetisation $D = 14°$ and $I = 10°$, but the bulk of the sediments, including the western part of the outcrop and all of the upper sequence, have a characteristic remanence ca. $D = 160°$, $I = 35°$, interpreted as a magnetic overprint during the Sveconorwegian mobile episode (1100–800 Ma). The lavas have low deuteric oxidation states but were subjected to widespread hematisation during later hydrothermal activity. The majority of the lavas, including the least altered, record a remanence with a mean $D = 4°$, $I = -2°$ interpreted as a primary field direction, and an antiparallel remanence locally present in hydrothermal hematite; a Sveconorwegian overprint has displaced this at some sites. Stratigraphic sections identify a polarity history N→R in the lower sediments followed by N→R in the lavas. The post-Jotnian dolerites ($D = 37°$ $I = -33°$) intruded at ca. 1250–1220 Ma did not achieve significant remagnetisation of the Jotnian rocks. Palaeomagnetic poles are at present insufficient to assign an age to the Jotnian rocks within the interval 1600–1250 Ma. □*Palaeomagnetism, pole positions, axis of magnetisation, reversals, magnetostratigraphy, petrography, hematite, lavas, sandstones, dolerites, Jotnian, post-Jotnian, Sveconorwegian, Proterozoic, Dala Group, Hedmark fylke, southeastern Norway, west central Dalarna, central Sweden, N6015 N6200 E1435 E1215.*

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The data of Precambrian palaeomagnetism are of two fundamentally different kinds: the first kind is derived from slowly-cooled plutonic igneous and metamorphic rocks which yield time averages of the geomagnetic field over periods of time usually exceeding 10^5 years and sometimes 10^7 years (Pullaiah et al. 1975; Morgan 1976). The second kind is derived both from minor igneous bodies intruded into cold crust and from volcanic and sedimentary successions. Under favourable circumstances these may yield essentially instantaneous records of the geomagnetic field and (in the latter case) place this information in a time sequence. Real progress in understanding the Precambrian geomagnetic field is hampered by the paucity of suitable volcanic and/or sedimentary suites. Studies to date on the ca. 1100 Ma Keweenawan lavas of North America (eg. Palmer 1970) and the ca. 1300 Ma Gardar lavas of Greenland (Piper 1977) have recovered a primary remanence and a sequence of magnetic field reversals. (All ages quoted in this paper have been recalculated to the decay constants listed by Steiger & Jäger [1977].) This paper reports a comparable study of the Jotnian lavas and sediments of western Sweden and eastern Norway.

Geology and age

The term 'Jotnian' (Sederholm 1927; von Eckermann 1937) is used to embrace the sedimentary and volcanic suites within the Fennoscandian Shield post-dating the last Precambrian mobile episode. It is now known (Magnusson 1960) that rocks assigned to this group actually predate the Sveconorwegian mobile episode (ca. 1100 Ma) in southwestern Scandinavia. Although sporadically distributed throughout the Fennoscandian–Ukrainian Shield and linked on a lithological basis, by far the largest outcrop is in the Dalarna district of western Sweden. This outcrop covers some 1200 km² straddling the Norway–Sweden border and is partially overthrust at its northwestern margin by Cambrian and younger rocks deformed within the Caledonian orogenic belt (Fig. 1); it is with this outcrop that the present study is concerned.

The Precambrian basement to this region com-

prises infracrustal and supracrustal rocks incorporated within the Svecofennian mobile belt and subjected to migmatisation at ca. 2000–1750 Ma (K–Ar and U–Pb determinations, Magnusson 1960; Welin & Blomqvist 1964). The termination of this episode is defined by the late to post-orogenic Revsund granite (Rb–Sr age of 1748±40 Ma, Welin et al. 1971) and the post-orogenic Värmland granite group (Rb–Sr age 1654±30 Ma, Welin 1970) Shortly afterwards extensive (sub-Jotnian) igneous activity took place. The products of this activity have been divided into the Lower and Upper Dala 'Series' by Hjelmqvist (1966); the former comprises rhyolites (porphyritic), quartzites and conglomerates, and the latter comprises rhyolites (non-porphyritic), ignimbrites and various clastic rocks. Both are extensively intruded by granites of which many have rapakivi characteristics. Welin et al. (1966) obtained an Rb–Sr age of 1650 Ma, and Priem et al. (1968) an Rb–Sr age of 1634 Ma on the Dala rocks. The Dala igneous province is thus nearly contemporaneous with the type-rapakivi magmatism of Finland and developed only shortly after stabilisation and uplift of the Svecofennian terrain and intrusion of the Värmland granites in the region; this close genetic relationship has prompted Lundqvist (1968) to regard the Dala rocks as anatectic magmas originating during the Svecofennian mobile episode.

Evidence of a major unconformity between the Dala Group and the overlying Jotnian is lacking (Lundqvist 1968). This latter sequence comprises a sedimentary–volcanic suite in excess of 800 m thickness and divided into a lower and upper part by some 100 m of lavas and tuffs (the 'Öje Diabase'). Deposition was controlled by SE–NW graben faulting and took place predominantly in shallow water under fluviatile conditions: ripple marks, mud cracks and rain drop impressions are preserved in the finer facies. Although conglomerates occur near the base, the sequence is predominantly a cross-bedded, pale red, quartzitic sandstone (Hjelmqvist 1966) comprising quartz, feldspar (usually subordinate) and clasts of Dala volcanics (locally); grains are thoroughly cemented by a matrix of quartz and feldspar or sericite making the sandstone facies very difficult to drill in the field. Red or chocolate shale horizons also occur sporadically within the succession.

The lava sequence (Fig. 1) comprises an unknown number of amygdaloidal and (sometimes) porphyritic olivine basalt flows, which are locally scoriaceous or pillowy (Hjelmqvist 1966). They are composed of an ophitic groundmass of labradorite or bytownite feldspar surrounding pseudomorphs of sericite, chlorite and ores after olivine and augite; vesicles are filled with hydrothermal epidote, calcite, chalcedony or chlorite. The Jotnian sequence is extensively cut by ('post-Jotnian') dolerites mostly in the form of sills and thick SE–NW dykes; these are fresh ophitic dolerites comprising labradorite feldspar, augite, olivine, ilmenite and magnetite. They have been the subject of extensive Rb–Sr studies by Patchett (1978) who has derived ages of 1232–1228 Ma (Idre), 1217 Ma (Mossiberg), 1247 Ma (Bornåssjön), 1231 Ma (Älvdalsåsen), 1223 Ma (Emådalen) and 1290–1215 Ma (Älvho); they have a comparable age and magnetisation to the extensive post-Jotnian dolerites of eastern Sweden and Finland (Patchett et al. 1978; Piper 1979).

The Jotnian succession is nearly flat-lying over most of the outcrop and forms a very broad syncline (Fig. 1) with structural dips of 15° or less; the exception is the SW margin where the outcrop abuts against upthrust basement rocks and structural dips are here locally near vertical (Hjelmqvist 1966).

It is not yet possible to assign a specific age to the Jotnian rocks within the interval 1600–1250 Ma. K–Ar ages fall in the range 1320–856 Ma (Magnusson 1960), but have apparently all been subject to argon loss during later tectonic–magmatic events. Other evidence is circumstantial: sediments assigned to the Jotnian at Nordingrå in east Sweden are older than 1385 Ma (Welin & Lundqvist 1975), and borehole sedimentary material from Gotska Sandön in the Baltic Sea north of Gotland and possibly Jotnian in age has yielded K–Ar ages of 1550 and 1500 Ma (Gorbatschev 1962).

Palaeomagnetic results

The sample

At each palaeomagnetic site (Fig. 1) samples were typically drilled over several metres of outcrop using a portable drilling motor of the kind described by Doell & Cox (1967). Cores were oriented by sights on the Sun, an altimeter was used to record the height of each site, and the structural dips were recorded wherever they could be reliably established; our dip determinations range from 23 to 0°. Unfortunately the Jotnian outcrop is extensively covered by glacial drift and exposure is very poor; it is thus not

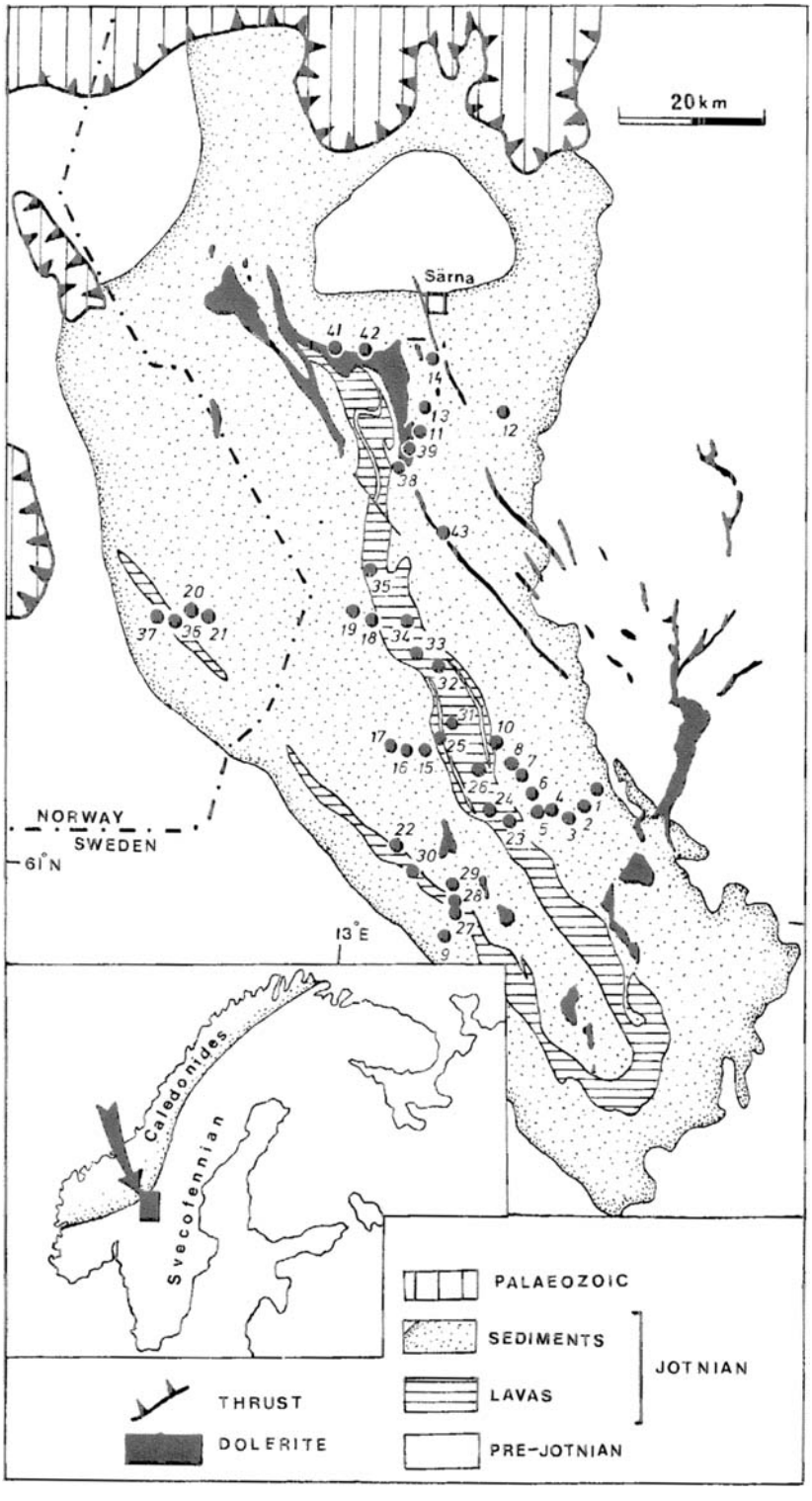


Fig. 1. Geology and palaeomagnetic sampling sites in the Jotnian lavas and sediments and post-Jotnian dolerites of central Scandinavia; the inset map shows the regional location.

possible to be confident of the stratigraphic succession of sampled sites except over very limited areas.

In the laboratory, sample magnetisations were measured with an automated parastatic magnetometer capable of accurate directional measurements on samples with a minimum intensity of about 0.02×10^{-5} A m^2 kg^{-1}. The lavas were progressively cleaned in alternating fields (a.f.) to peak fields ranging from 50 to 150 mT, and were then progressively thermally cleaned in a range of fields up to 610 or 640°C. The sediments were unaffected by a.f. demagnetisation and were accordingly only cleaned by the thermal method at a range of increasing temperatures described below. The oven used was a commercial Schonstedt TSD-1 type with axial fields nullified to less than ± 20 nT.

The Jotnian sediments

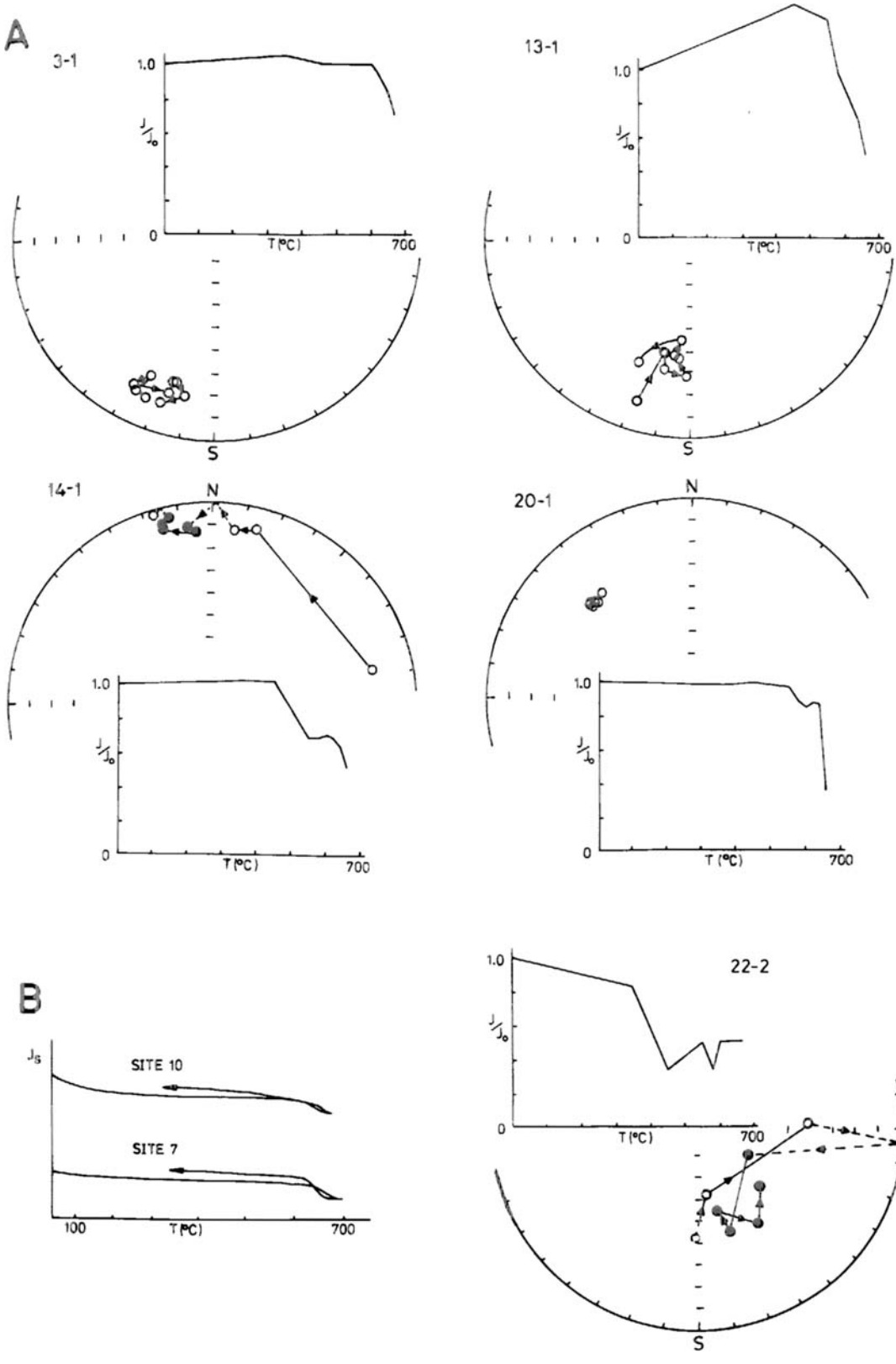
The 22 sites in the Jotnian sediments can be conveniently subdivided into those sites (1 to 14) derived from the lower part below the lavas and those sites (15 to 22) from the upper part above these lavas (Fig. 1). One pilot core from each site was treated at successive temperatures of 350, 450, 550, 580, 600, 620, 640 and 660°C and the remaining samples were treated at 3 or 4 blanket temperatures selected from the behaviour of the pilot. All directions discussed here are corrected for tectonic tilts. The sediments possessed total NRM intensities in the range $0.40\text{–}0.02 \times 10^{-5}$ A m^2 kg^{-1}. The thermal decay curves (eg. Fig. 2) exhibit, in the majority of cases, high blocking temperature spectra indicating that hematite ($T_c \sim 670$°C) is the predominant remanence carrier; and directions randomise after significant fall in remanence at higher temperatures (Fig. 3). Thermomagnetic determinations (saturation magnetisation, J_s, versus temperature) also confirm that hematite is the major remanence carrier: weak initial saturation remanence values fall to zero close to the Curie point of pure hematite (Fig. 2).

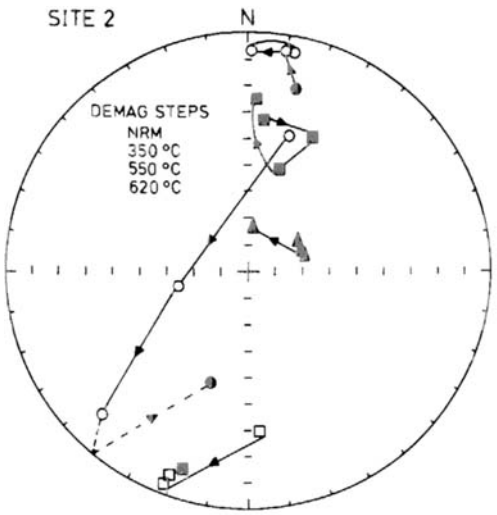
Many samples exhibit no significant change in direction with this progressive treatment and directions defined by thermal treatment near the Curie point are similar to those at the total NRM level, but there are some exceptions such as site 14 (Figs. 2–3) where the progressive thermal treatment produces a large initial movement to a northerly shallow negative direction. A further change in inclination from shallow negative to shallow positive between 450 and 550°C produces a further fall in intensity from the secondary plateau, and is indicative of two high blocking temperature components residing in magnetite and hematite. Thermal cleaning of some other samples leads to removal of variable low blocking temperature components (Fig. 3). The quality with which stable end points can be estimated is a function of the residual intensity: the direction defined by high temperature demagnetisation of sample 22-2 for example (Fig. 2) corresponds to a plateau on the thermal decay curve equivalent to an intensity of 0.03×10^{-5} A m^2 kg^{-1}, whereas the end point of sample 13-1 is better defined by residual intensities in the range $0.14\text{–}0.08 \times 10^{-5}$ A m^2 kg^{-1} (Fig 2).

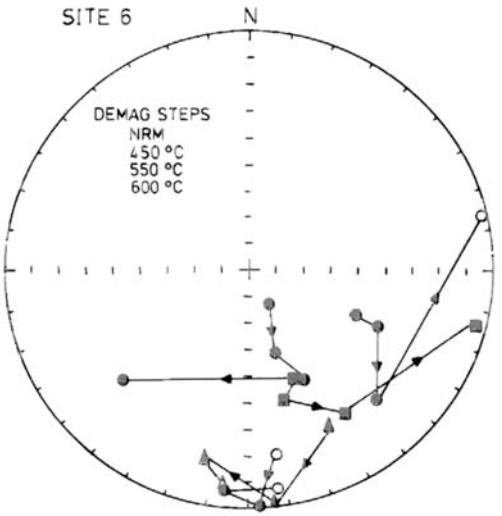
The site mean directions before and after cleaning are listed in Table 1; these are not confined to a common direction but six sites exhibit antiparallel shallow positive northerly and shallow negative southerly directions (Fig. 4) while the remainder yield directions intermediate between the two; the significance of these directions is considered in the next section. Sites 2 and 12 (Fig. 3) clearly show two distinct groups of mean directions. At site 2 these are closely antiparallel and one group has been reversed to derive the group mean. At site 12 the sample directions are not antipallel and both a northerly shallow positive and a SW steep positive direction are represented. At site 6 three of the samples move towards a common mean, but intensities drop below the limits of detectability above 600°C and we could not confirm that it had reached a stable end point. Six of the 22 sites have been rejected because the within-site scatter was large and α_{95} exceeded 40°.

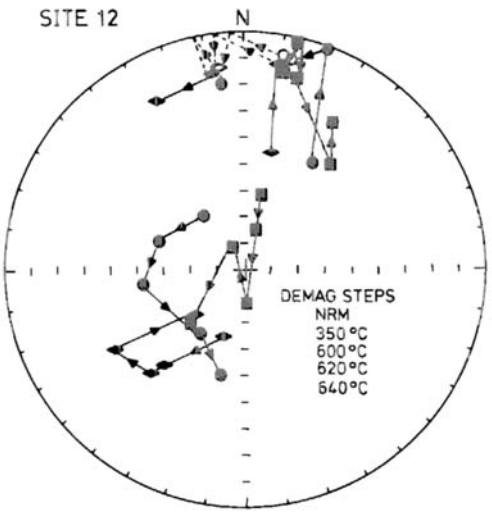
The distribution of site mean directions shows two antiparallel groups of three sites (Fig. 4) defining the axis of magnetisation $D = 14°$, $I = 10°$ (6 sites, $\alpha_{95} = 13°$) corresponding to a palaeomagnetic pole at 177°E, 33°N. All sites defining this axis are present in the lower part of the sequence and the stratigraphic succession is known at two localities where the sites are in the order of increasing stratigraphic height $2 \rightarrow 3 \rightarrow 4 \rightarrow 6 \rightarrow 8$ and $12A \rightarrow 14 \rightarrow 13$ respectively. Both of these sequences define a complete polarity transition represented by a movement from

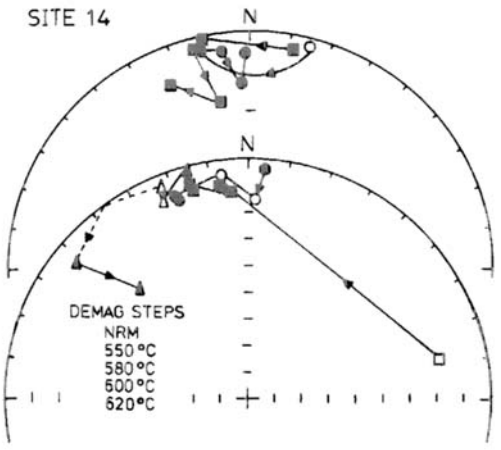
Fig. 2. □ A. Behaviour of directions of magnetisation and magnetic moments of Jotnian sediment samples with progressive thermal cleaning. Open symbols are upper hemisphere plots and closed symbols are lower hemisphere plots, and moments are expressed as a fraction of the initial moment J_0: stereographic projections. □ B. Thermomagnetic determinations (saturation magnetisation J_s, versus temperature in °C) for two sediment samples; the J_s axis is arbitrary.

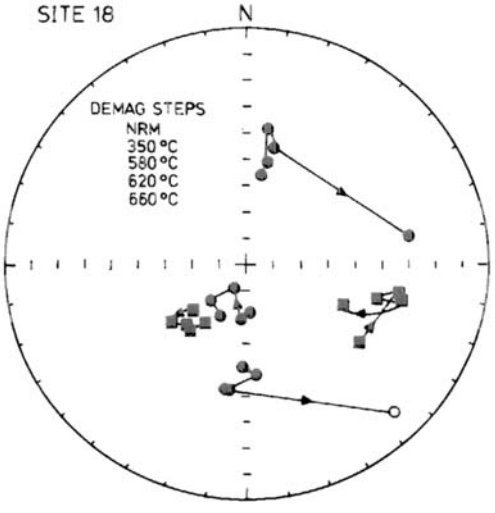

A
3-1
1.0
J/J₀
0
T(°C)
700
S
13-1
1.0
J/J₀
0
T(°C)
700
S
14-1
N
1.0
J/J₀
0
T(°C)
700
20-1
N
1.0
J/J₀
0
T(°C)
700
B
Js
SITE 10
SITE 7
100
T(°C)
700
1.0
J/J₀
0
T(°C)
700
22-2
S


SITE 2
N
DEMAG STEPS
NRM
350 °C
550 °C
620 °C


SITE 6
N
DEMAG STEPS
NRM
450 °C
550 °C
600 °C


SITE 12
N
DEMAG STEPS
NRM
350 °C
600 °C
620 °C
640 °C


SITE 14
N
N
DEMAG STEPS
NRM
550 °C
580 °C
600 °C
620 °C


SITE 18
N
DEMAG STEPS
NRM
350 °C
580 °C
620 °C
660 °C

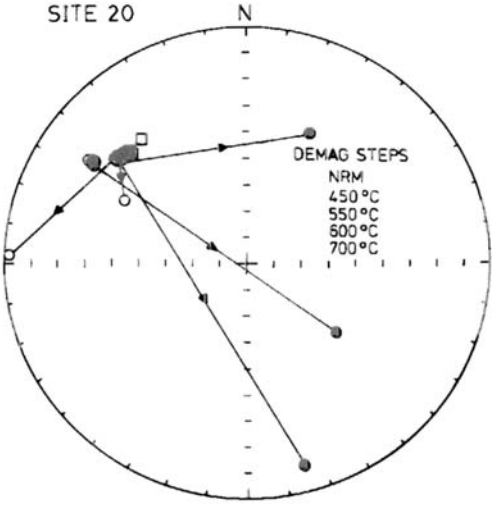

SITE 20
N
DEMAG STEPS
NRM
450 °C
550 °C
600 °C
700 °C

Table 1. Palaeomagnetic results for Jotnian sediments, (lower part: sites 1–14; upper part: sites 15–22) site and group mean data after tilt correction and before and after thermal cleaning. R is the resultant vector of N samples with Fisher precision parameter k [$= (N-1)/(N-R)$], and mean direction with declination D and inclination I; α_{95} is the radius of the circle of 95% confidence about the mean direction in degrees; dp and dm are the radii of the 95% confidence oval about the palaeomagnetic pole in the colatitude direction and at right angles to it respectively.

Before cleaning						After cleaning					
Site No.	N	R	k	D	I	N	R	k	α_{95}	D	I
1	5	–	–	–	–	–	–	–	–	–	–
2	3	2.92	28.0	2.6	+27.4	4	3.74	11.5	28.3	22.8	+12.0
3	5	4.36	6.3	204.5	+ 1.3	5	4.54	8.6	27.6	202.9	−11.5
4	3	2.76	8.2	214.7	−20.7	3	2.86	14.4	33.8	205.3	− 3.6
5	5	–	–	–	–	–	–	–	–	–	–
6	4	3.74	11.4	142.7	+49.8	5	4.58	9.6	26.0	162.7	+21.6
7	2	1.98	44.5	142.0	+29.3	4	3.65	8.6	33.3	154.1	+26.3
8	2	1.99	107.9	167.2	+45.1	4	3.78	13.8	25.6	159.4	−26.3
9	7	–	–	–	–	–	–	–	–	–	–
10	5	4.91	44.2	8.7	+75.4	–	–	–	–	–	–
11	5	–	–	–	–	–	–	–	–	–	–
12	3	2.96	45.4	20.2	+50.2	(A) 3	2.96	56.7	16.5	12.1	+ 3.6
						(B) 3	2.95	37.4	20.4	216.6	+52.7
13	7	6.89	55.2	196.1	− 8.7	7	6.93	89.8	6.4	193.6	−23.4
14	3	2.98	109.4	9.9	+ 2.2	5	4.93	60.8	9.9	347.3	+ 4.5
15	2	1.97	40.7	167.0	+38.8	3	2.95	44.9	18.6	157.1	+29.0
16	4	3.75	12.2	35.0	+59.3	5	4.65	11.3	23.8	75.6	+66.2
17	4	3.69	9.6	172.6	+ 8.4	–	–	–	–	–	–
18	3	2.92	24.5	194.3	+63.9	3	2.91	21.2	27.5	201.5	+59.5
19	4	3.46	5.5	249.3	+70.5	4	3.81	16.2	23.6	159.4	+45.6
20	6	5.97	192.1	311.2	−27.6	6	5.98	260.3	4.2	310.0	−27.9
21	3	2.97	74.2	328.4	+64.6	3	2.94	32.7	21.9	304.4	+68.8
22	4	–	–	–	–	4	3.84	19.2	21.5	149.5	+47.9

Mean direction and palaeomagnetic pole (sites 2, 3, 4, 12A, 13, and 14)

R	D	I	k	α_{95}	Palaeomagnetic pole
5.81	14.1	10.0	26	13.4	176.5°E, 32.6°N ($dp = 6.9°$, $dm = 13.5°$)

northerly to southerly declinations. Since the sampled sections are on a direct strike continuation of one another (Fig. 1) we conclude that they represent the same geomagnetic event.

The remaining sites define intermediate directions of magnetisation (>40° from the mean dipolar axis), and these are the only directions present in the upper part. The most prominent feature of these directions is a group ('B') with south easterly declinations and moderate positive inclinations; five sites falling in this group (6, 7, 15, 19 and 22) yield a group mean of $D = 157°$, $I = 34°$ ($\alpha_{95} = 12°$) corresponding to a palaeomagnetic pole at 35.3°E, 7.7°S, and site 20 appears to be antiparallel to this group. Unfortunately the absence of potential field tests and the advanced lithification of these rocks preclude statistical or chemical tests, but we have identified no clear differences in the magnetic properties of the sites yielding these two distinct sets of directions.

Fig. 3. Behaviour of sample directions of magnetisation with progressive thermal treatment at several sites in the Jotnian sediments; cleaning stages are indicated and symbols are the same as for Fig. 2.

The lavas

The two stage (a.f. followed by thermal) treatment was found to be essential to the complete evaluation of the directional behaviour in many of these samples, and some typical examples are illustrated in Fig. 5. Sites 23 and 24 however, showed no further change with progressive thermal treatment while site 24 possessed a soft component similar to the primary remanence of the post-Jotnian dolerites (Table 2) and readily removed by a.f. cleaning. Site 25 possessed a steep positive component only successfully re-

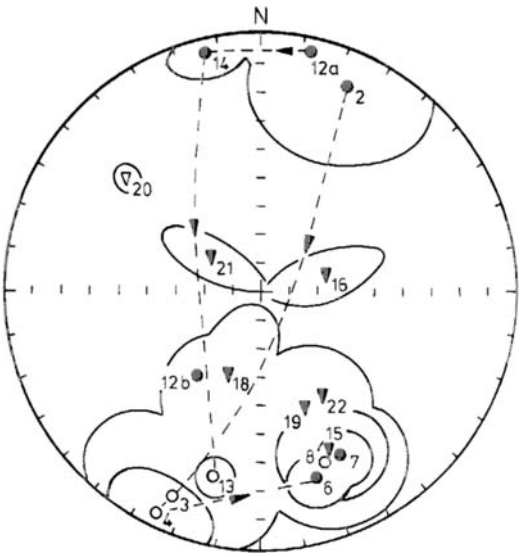


Fig. 4. Site mean directions of magnetisation with 95% confidence circles from the Jotnian sediments; the arrowed lines connect sites in stratigraphic order where this is known and sites in the lower part are indicated by circles and sites in the upper part by triangles. Stereographic projection.

moved by a.f. cleaning in the lower coercivity samples; the remainder moved to shallow northerly directions with progressive thermal cleaning and two samples exhibited a complete reversal of polarity above 545°C (Fig. 5). The complex magnetic record at this site would appear to reside in components of magnetite and hematite both of which are identified in petrographic and thermomagnetic analysis (Table 3). Site 27 possessed very high coercivity remanence with no change in moment until the Curie point of magnetite was reached; hematite is also clearly detected in thermomagnetic spectra (Table 3). Like Site 26, thermal treatment has little effect on the direction of remanence except for a migration of a few degrees towards the primitive suggesting that a.f. cleaning has not been entirely successful in removing a small component possibly in the present earth's field direction; the thermally-cleaned results are therefore accepted in Table 2.

Site 28 possessed weak moments with directions which survived thermal cleaning with a slight reduction in inclination. A comparable direction was defined by a.f. cleaning of three of five samples of Site 29 (Table 2); with thermal cleaning above 500°C however, these samples retain an approximate direction, $D = 180°$, $I = 50°$ but disperse again above 600°C so that an end point could not be accurately estimated. At Site 30 a direction $D = 60°$, $I = 54°$ defined by a.f. cleaning did not survive thermal treatment and the mean direction migrated towards a northerly shallow direction (Fig. 5, Table 2). Sites 31 and 32 were stable to progressive a.f. and thermal treatment and no residual component remained above the Curie point of magnetite. Site 33 was also very stable but the mean defined by a.f. treatment has a similar declination (but shallower inclination) to the post-Jotnian dolerites. That a small post-Jotnian secondary component remained here, was further suggested by thermal demagnetisation which produced small movements of a few degrees towards more northerly declinations.

Two components of magnetisation were identified at Site 35 where both magnetite and hematite were detected in thermomagnetic spectra (Table 3); with a.f. treatmen all samples moved towards a mean $D = 111°$, $I = 52°$ but with progressive thermal cleaning further movement towards shallow inclinations was identified in four samples (Fig. 5) yielding a final mean of $D = 128°$, $I = -2°$ (Table 2). Magnetite and hematite were also identified at site 36 where three samples showed a complete reversal of direction of magnetisation above 545°C (Fig. 5) and a small residual component remaining above 600°C; these sample directions have been reversed to calculate the group mean of Table 2. At Site 37 a single component is stable to the Curie point of magnetite; no hematite component was identified at this site.

Thus, with comprehensive treatment, several components of magnetisation are identified at most sites, and these are normally where hematite is specifically identified in the thermomagnetic determinations (Table 3). Stable end points are defined for the majority of sites but there are a few cases (Sites 29, 30 33 and 35) where samples are either poorly stable or continue to show systematic movement with progressive treatment right up to the Curie point(s), and in these instances the final end points are not considered to have been accurately defined.

Some individual samples with multicomponent magnetisations define a dipolar axis of magnetisation which is also identified by the final group means of the majority of sites. These sites (25, 26, 27, 30, 31, 32, 34, and 36) yield a group mean direction of $D = 4°$, $I = -2°$ which is similar to the dipolar axis of magnetisation defined from six of the sites in the sediments (Table 1). While the magnetite-held component in these lavas may represent a magnetisation acquired during deuteric processes at initial igneous cool-

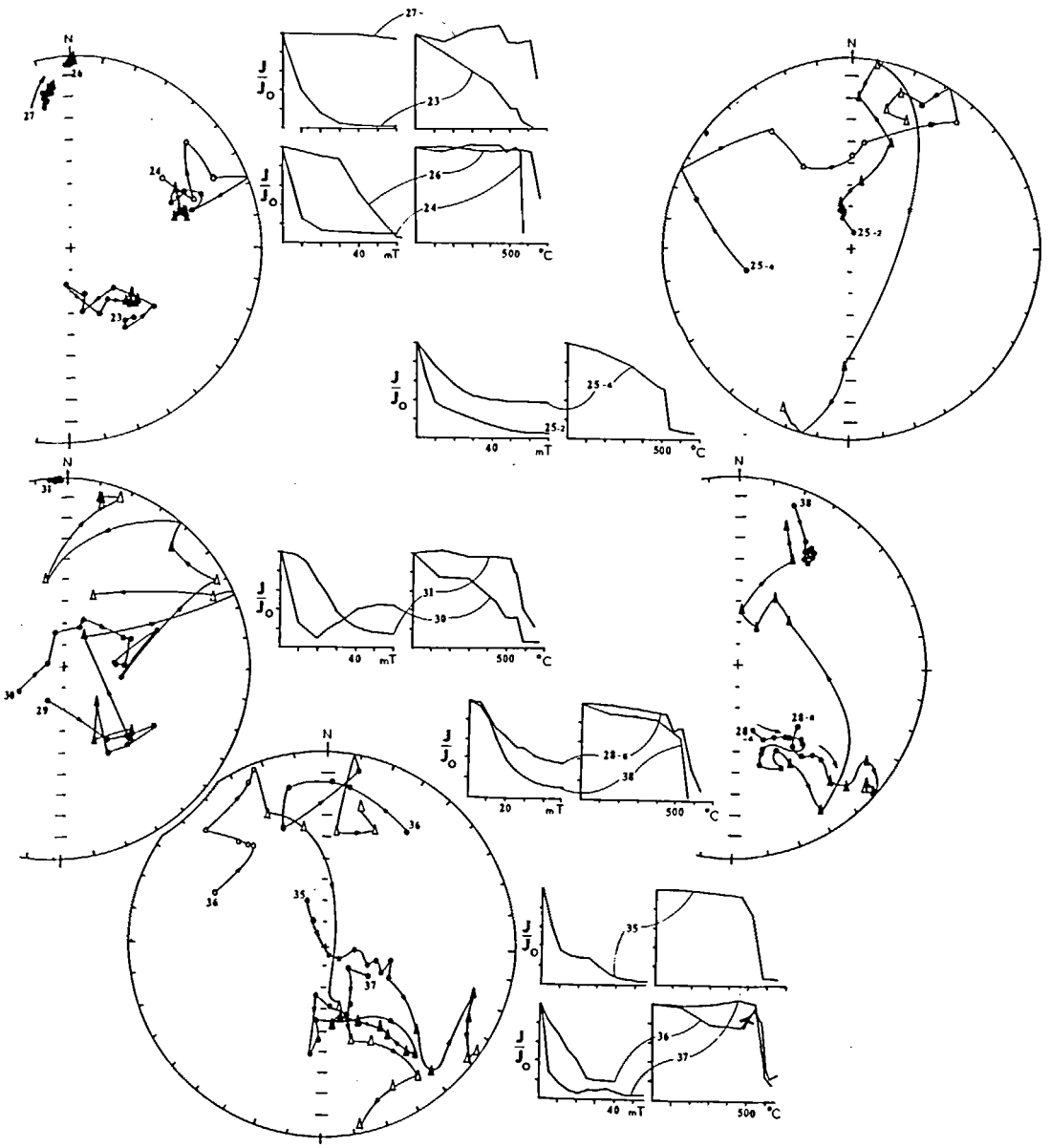


Fig. 5. Behaviour of directions of magnetisation of samples of Jotnian lavas and post-Jotnian dolerites with progressive a.f. (circles) and thermal (triangles) treatment; magnetic moments are expressed as a fraction of the initial moment J_0 and other symbols are as for Fig. 2. For clarity only some demagnetisation stages are illustrated here.

ing, it is probable that the reversed (shallow southerly) component residing in the hematite was acquired by hydrothermal or lithification processes (Table 2) over a longer period of time; in this context it may be significant that no intermediate direction of magnetisation is preserved in these multicomponent samples.

The remaining eight sites yield directions of magnetisation (>40° from the mean) of which four (23, 24, 28, and 37) are considered to be well defined by the demagnetisation treatment; they are not, however, collectively well grouped. The stratigraphic succession of sampling sites is known in three areas (Fig. 1); these are in order of increasing stratigraphic height 26 → 24, 27 → 28 → 29, and 32 → 33 → 35, and there appears to be no systematic migration away from a shallow northerly direction (the dipolar axis) to any

Table 2. Palaeomagnetic results for Jotnian lavas and post-Jotnian dolerites after a.f. and thermal cleaning and tilt correction.

Site No.	*N*	*D*	*I*	*R*	*k*	α_{95}	J_0($\times 10^{-5}$ A m^2 kg^{-1})	Curie point(s) (± 10°C)	Curie point curve type
Jotnian lavas									
23[a]	6	136.0	65.1	5.95	111	6.4			
24[a]	4	68.8	24.7	3.97	87	9.9			
25	7	6.3	−12.4	6.42	10	19.4			
26	6	1.1	1.2	5.99	1006	2.1			
27	8	348.0	16.5	7.96	163	4.4			
28[a]	5	137.4	25.7	4.86	29	14.4			
29[a]	(I) 5	179.2	52.4	4.56	9	26.7			
30	5	18.3	1.0	4.29	6	35.5			
31	5	0.6	−6.7	4.91	43	11.8			
32	4	20.2	8.3	3.77	13	26.5			
33[a]	7	53.0	−1.3	6.88	52	8.5			
34	5	354.9	−10.9	4.85	27	15.1			
35[a]	(A)7	111.1	51.6	6.26	8	22.5			
	(B)4	127.7	−2.4	3.84	19	21.4			
36	6	2.0	−14.5	5.74	19	15.7			
37[a]	6	174.8	51.6	5.97	159	5.3			
Post-Jotnian dolerites									
38	5	31.4	−30.4	4.98	238	4.9	3.60–1.80	570	A
39	6	36.7	−27.8	5.99	645	2.6	1.89–0.51	570	A
40	6	38.5	−34.1	5.89	46	9.9	6.45–1.00	565	A
41	5	38.7	−36.3	4.92	49	11.1	0.87–0.08	555	A
42	5	37.4	−37.0	4.98	262	4.7	9.28–0.18	560	A
43[a]	6	0.4	82.2	5.99	459	3.1	3.23–2.10	565	A
Group mean directions and palaeomagnetic poles								Palaeomagnetic pole	
Jotnian lavas:	8	4.0	−2.2	7.75	28	10.7		188.7°E, 27.3°N ($dp = 5.3°$, $dm = 10.7°$)	
Post-Jotnian dolerites:	5	36.5	−33.1	4.99	301	4.4		158.6°E, 5.3°N ($dp = 2.8°$, $dm = 5.0°$)	

[a] Sites excluding from group mean calculations. Other symbols as for Table 1 and Table 3.

single intermediate direction (Fig. 6). We conclude that these estimates of intermediate field directions are mostly likely to represent remagnetisation at some later time and six of them (23, 28, 29, 35A–B, and 37) are comparable to intermediate directions from the sediments.

The post-Jotnian dolerites

Post-Jotnian dykes and sills cutting the Jotnian sedimentary volcanic succession were collected at six localities (Fig. 1). By way of contrast to the lavas they exhibit very simple behaviour to progressive cleaning treatment: initial directions with shallow magnetic inclination exhibit slight initial increase in magnetic moments and migrate towards directions with slightly greater negative inclination during progressive a.f. treatment. Median destructive fields are in the range 20–50 mT which yield stable end points showing no further migration with progressive thermal treatment; moments fall to zero close to the Curie point of pure titanomagnetite and thermomagnetic determinations on these dolerites (Table 2) yield single comparable Curie points (555–570°C) and reversible spectra suggesting that titanomagnetite is the only remanence carrier

All sites (with the exception of 43) yield NE negative directions of magnetisation with a mean of $D = 37°$ $I = -33°$ ($\alpha_{95} = 4.4°$), and a palaeomagnetic pole at 159°E, 5°N; interpretation of this pole is considered in the next section.

Magnetic-petrologic properties of the lavas

Lava successions of Proterozoic age (including the Jotnian) typically possess associations of secondary minerals, and these were usually pro-

Table 3. Summary of magnetic-petrologic properties of the Jotnian lavas. J_0 is the total NRM intensity of magnetisation. Curie point curves are defined as: A, heating and cooling curves are similar; B, heating curve below cooling curve; C, heating curve above cooling curve; D, very large increase in saturation remanence on cooling. The scatter expresses the ranges of magnetite oxidation as $\sum_{n=1}^{6}$ (mean oxidation class $-n$) × area of n/ $\sum_{n=1}^{6}$ area

Site No.	J_0 ($\times 10^{-5}$ A m^2 kg^{-1})	Curie point(s) (± 10°C)	Curve type	Titanomagnetite		Hematite	Sulphides (%)
				Oxidation class (scatter)	Volume (%)		
23	8.69–1.55	555,650	A	3.6 (0.90)	7.0	Present[a]	–
24	0.23–0.02	575	D	1.2 (1.31)	3.5	Present[a]	–
25	1.60–0.32	580, 630	A	1.5 (0.59)	8.2	Present[b]	–
26	0.96–0.09	575, 660	A	1.1 (0.16)	0.5	Present[a]	–
27	9.37–0.47	575, 650	D	1.1 (0.12)	1.5	Present[a]	–
28	0.35–0.01	550	C	1.0	1.6	Present[b]	–
29	38.08–2.03	565, 630	A	1.5 (0.69)	9.8	Present[b]	–
30	1.60–0.04	555, 670	A	1.3 (0.54)	12.0	Present[b]	–
31	5.56–0.47	565	A	1.2 (0.30)	0.1	–	0.2
32	0.20–0.11	575	D	1.7 (0.66)	0.5	–	
33	0.88–0.41	575	A	1.0	0.4	–	0.1
34	2.39–0.90	560	A	1.2 (0.45)	9.3	–	2.4
35	1.44–0.13	570, 620	C	2.0 (0.44)	9.9	Present[b]	–
36	0.04–0.01	570, 660	C	1.0	21.9	Present[b]	–
37	2.23–0.95	570, 670	A	1.2 (0.25)	7.4	Present[b]	–

[a] Disseminated and abundant in grain sizes < 2 μm down to limits of resolution.
[b] Present in more granular form than 1, up to 5 μm in diameter.

duced by hydrothermal alteration under conditions indicating higher temperature gradients than those which have influenced Phanerozoic volcanic successions (Ade-Hall et al. 1971). An opaque petrological assessment is necessary to evaluate the extent to which the magnetic constituents have been affected by this low grade metamorphism, and hence to give a corroborative assessment of any remagnetisation effects.

For petrographic analysis a minimum of 50 grains were measured and classified according to the scheme of Wilson & Watkins (1967) and Ade-Hall et al. (1968) and the practical results were analysed by computer as described in Table 3. Grains were assigned a class number according to the degree of sub-solidus oxidation of high-Ti titanomagnetite to low-Ti titanomagnetite and ilmenite, and progressive oxidation of ilmenite lamellae; the mean class of all grains gives the oxidation number. The classes (described in detail in Ade-Hall et al. 1968) are: (1) unoxidised; (2) less than 50% of the grain covered by ilmenite lamellae; (3) more than 50% of the grain covered by ilmenite lamellae; (4) ilmenite lamellae pseudomorphed by titanohematite and ferri-rutile; (5) pseudomorphed ilmenite lamellae with spinel exsolution rods in the magnetite; (6), primary grain entirely replaced by an aggregate of titanohematite, pseudobrookite and ferri-rutile. All of these phases can be formed under appropriate conditions of deuteric oxidation following initial cooling over the temperature range ca. 750–400°C, but it is difficult or impossible to isolate the effects of hydrothermal alteration (< 350°C) in the more oxidised phases because titanohematite and ferri-rutile can be produced by both processes.

Epidote is visible in some Jotnian lavas implying that hydrothermal temperatures of 300–400°C (Ade-Hall et al. 1971) were reached at least locally. Hematisation of magnetite and relict ilmenite in a base of titanohematite and ferri-rutile is characteristic of hydrothermal alteration to this grade (Ade-Hall et al. 1971; Piper 1977) and is extensively developed in more than one half of this collection (Table 3); in this case it is certainly of hydrothermal origin because it has affected lavas with low deuteric oxidation states. In general the class of deuteric oxidation is low in these lavas and falls mostly between 1 and 2 (Table 3). With one exception, the hematite is only detected in thermomagnetic spectra where it occurs in distinct granular form, and it is these samples which have the most complex magnetic record (Table 3) it has not in general been detected where it is present as the fine disseminations with grain sizes of 2 μm or less. At sites 31 to 34 (from the central part of the eastern limb (Fig. 1)) the effects of hydrothermal alteration have been minimal and primary sulphides are

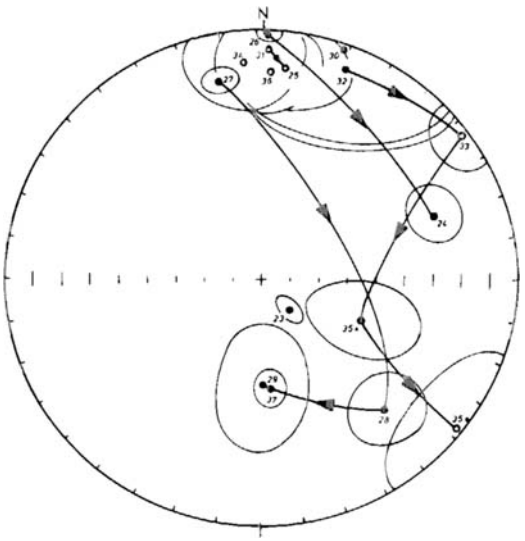


Fig. 6. Site mean directions of magnetisation from the Jotnian lavas with the stratigraphic succession of sites (where known) indicated by arrowed lines; 95% confidence circles are also given. Stereographic projection.

preserved; it is these sites which possess a magnetic record which is simple, or relatively so, and they are included within the coherent group of sites exhibiting northerly shallow directions. The intermediate directions however, appear to exhibit no distinctive petrographic properties to separate them from the remaining sites (Table 3).

Discussion of results

Six sites from Jotnian hematite-bearing sediments of this study define a dipolar axis of magnetisation $D = 14°$, $I = 10°$, and eight sites from the lavas define a mean $D = 4°$, $I = -2°$. We believe that these directions define the geomagnetic field axes at the time of deuteric–hydrothermal alteration of the lavas and deposition–lithification of the sediments, because (1) they include directions of both polarities, (2) it is the only coherent direction recovered from the lavas, and (3) it includes the lavas subjected to minimal alteration. In addition, these directions reside in hematite at some sites and in magnetite at others.

The intermediate direction might in part be explained in the context of transitional field directions, but this is unlikely to be a significant cause both because of the large number of directions represented and because they reside in magnetic moments of comparable magnitude to the remaining sites. Shaw (1975) demonstrated that high field intensities may be attained during a polarity transition but they are only briefly achieved when the pole moves to equatorial latitudes. In this case we have a field direction at about 45° to the main dipolar field. The mean computed from sites 6, 7, 15, 19, 22, 23, 28, 29, and 37 is $D = 157°$, $I = 41°$, $R = 8.6°$, with a palaeomagnetic pole at 35°E, 3°S ($dp = 8.8$, $dm = 14.5°$).

The most plausible explanation would appear to be remagnetisation, notably during the Sveconorwegian tectonic activity and subsequent uplift (1100–800 Ma). The western margin of the Jotnian outcrop was affected by these events and it is in the sediments of the upper part of the sequence in this region that the bulk of the intermediate directions are observed (most of the lava intermediate sites are also found here; Fig. 1). In their study of ca. 1010–890 Ma dolerite dykes parallel to the Sveconorwegian Front Zone, Patchett & Bylund (1977) isolate remanence directions of both polarities comparable to the 'B' axis identified in the Jotnian sediments. Comparable uplift magnetisations are also present within the Sveconorwegian belt (Poorter 1975) and the palaeomagnetic pole for the 'B' directions correlates closely with several listed by Patchett & Bylund in their Table 3; these latter authors examine in detail the tectonic significance of these palaeomagnetic poles.

It is noteworthy that remagnetisation by the post-Jotnian dolerite activity at 1200 Ma can be completely ruled out because these directions of magnetisation are quite different from the post-Jotnian dolerites. Although remagnetisation effects of these widespread dolerites have been identified at a few localities they are localised and minor. It thus seems likely that the dolerite intrusion was restricted to a short interval within the assigned age limits (ca. 1270–1220 Ma). Furthermore, the dolerites have an identical magnetisation to the post-Jotnian dolerites in eastern Sweden which are remote from both the Caledonian and Sveconorwegian Fronts and with which they are contemporaneous (Patchett 1978; Patchett et al. 1978; Larson & Magnusson 1976; Piper 1979); at least five of the six sites sampled in this study do not appear to have been affected by the Sveconorwegian activity.

The Sveconorwegian overprint in the Jotnian succession was probably achieved by near-surface processes because it is confined to the higher stratigraphic levels, and lower levels even near the Sveconorwegian margin (sites 27, 30, and 36) appear to retain primary remanence.

Table 4. Ca. 1650–1200 Ma palaeomagnetic poles from the Fennoscandian Shield.

Pole No.	Rock unit	Age (Ma)	Palaeomagnetic pole °E	°N	*dp/dm*	Reference
1	Åva intrusives, Finland	1700–1670	169	41	9/17	Neuvonen (1970)
2	Kumlinge dolerites, Finland	>1620	204	8	18/28	Neuvonen & Grundström (1969)
3	Föglö dolerite, Finland	≥1620	187	31	6/12	Neuvonen & Grundström (1969)
4	Upper-Dala volcanics, Sweden	<1634	176	23	–/–	Mulder (1971)
5	Loftahammar gabbro, Sweden	1660–1470	179	23	2/5	Poorter (1976)
6	Jotnian (Dalarna) sediments	–	177	33	7/14	This paper
7	Jotnian (Dalarna) lavas	–	189	27	5/11	This paper
8	Jotnian (Satakunta) sediments, Finland	ca. 1300	180	3	8/12	Neuvonen (1973)
9	Gävle granite, Sweden	>1250	165	35	7/14	Piper (1980)
10	Nordingrå gabbro-anorthosite, Sweden	>1385	138	37	4/6	Piper (1980)
11	Nordingrå granite, Sweden	1385	149	33	8/14	Piper (1980)
12	Ragunda complex, Sweden	1293	165	54	6/9	Piper (1980)
13	Vaasa dolerites, Finland	1270–1225	164	7	3/6	Neuvonen (1966)
14	Satakunta dolerites, Finland	1236–1225	158	2	3/5	Neuvonen (1965)
15	Märket dolerites, Finland	1270	146	−6	7/11	Neuvonen & Grundström (1969)
16	Post-Jotnian dolerites, E. Sweden[a]	1254	157	−7	2/3	Piper (1979)
17	Post-Jotnian dolerites, W. Sweden (1)	1250–1215	159	5	3/5	This paper
18	Post-Jotnian dolerites, W. Sweden (2)[b]	1250–1215	149	−1	–/–	Patchett et al. (1978)
19	Post-Jotnian dolerites, W. Sweden (3)[b]	1250–1215	147	3	–/–	Patchett et al. (1978)

[a] Mean of several studies, see reference.
[b] Virtual geomagnetic poles.

Mulder (1971) also investigated the magnetisation of Jotnian lavas from 9 localities and reported directions similar to the present field at 7 of these sites; two sites however, yielded a mean of $D = 7°$, $I = +6°$, similar to lava sites 25–27, 30–32, 34, and 36 of the present study. Noting that the pole position was comparable to Palaeozoic poles from Europe, Mulder concluded that either the field directions were similar during the Palaeoxoic and Jotnian interval of Precambrian times, or that remagnetisation took place in Palaeozoic times; he favoured the latter explanation. However, for the reasons given above we now regard this direction as primary and representative of the time of formation of the Jotnian volcanic-sedimentary succession.

Table 4 summarises the available palaeomagnetic poles for the Fennoscandian Shield relevant to the interval ca. 1650–1200 Ma. There are 19 data points and it is unlikely that they are sufficient to define the complete apparent polar wander (a.p.w.) path relative to the Shield for this interval; poles 13 to 19 for example, all relate to the post-Jotnian dolerites of Finland and Sweden and may represent a relatively short part of this time interval. The poles 6 and 7 for the Jotnian lavas and sediments fall close to three other poles (1, 3 and 5) dated about 1600 Ma; a significantly younger age between 1290 and 1220 Ma (Fig. 7) cannot, however, be ruled out. The Jotnian lavas and sediments of Dalarna yield quite different poles from the Jotnian Satakunta sediments of Finland (Fig. 7) emphasising that the sediments within the Fennoscandian Shield collectively grouped together as Jotnian are unlikely to be of the same age. Finally, pole 17 from the post-Jotnian dolerites of this region agrees closely with virtual geomagnetic poles from two sites reported by Patchett et al. (1978) and with poles from other post-Jotnian dolerites in eastern Sweden and Finland (Fig. 7).

A generalised a.p.w. swathe for the interval covered by the data of Table 4 is also given in Fig. 7. This commences with the ca. 1450 Ma pole 8 for the Satakunta Sandstone, passes through the ca. 1390 Ma Nordingrå poles and the 1290 Ma Ragunda pole, and terminates as a closed loop with the ca. 1270–1210 Ma post-Jotnian dolerite poles. It is probable that a considerable time interval is unrepresented here between the ca. 1600 Ma poles 1–7 and pole 8, and we are not, at present, justified in connecting them.

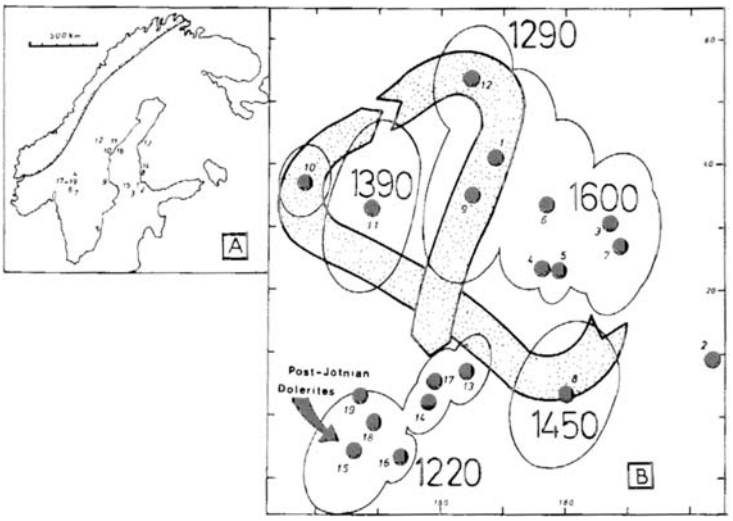


Fig. 7. □ A. Location of palaeomagnetic studies from the Fennoscandian Shield relating to the interval ca. 1650–1200 Ma. □ B. The poles are plotted as listed in Table 4 with 95% confidence ovals. The swathe is a generalised a.p.w. curve for the interval 1450–1200 Ma connecting the poles relating to this interval in order of known age; present data are inadequate to define the a.p.w. curve prior to ca. 1450 Ma. The large figures give the approximate ages in Ma.

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